

AI-Supported Planning — building the PoC in three weeks

The architecture for Step 1, and a day-by-day plan to build it. Scope is taken from the Requirements Ledger, which governs; the Dev Team Briefing supplies background.

Client Newvision Software GmbH **By** PerfektWerk **Build window** 3 weeks · 15 working days
Status Plan – pre-contractual PoC

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SECTION 1

What we are building

A working planning system for five P&L categories at total-company level, in which every number can be traced to its derivation on the spot. The Excel reference model already proves the mechanism; this PoC turns it into something that recalculates live, holds its own audit trail, and can survive a sceptical executive asking "where did that come from?" about any figure on screen.

IN SCOPE – LOCKED	EXPLICITLY NOT NOW – STEP 2
Five categories: Revenue, Material costs, External services, Personnel costs, Other costs (incl. Depreciation)	HR Planning, Resource Planning, Portfolio Planning
Total company level only	The full chart of accounts (~250–300)
Fixed + Variable split derived by regression from actuals	Departmental breakdown
Five-year plan, fully formula-driven	Cash-forecast / customer-payment-behaviour pilot
Best / Base / Worst scenarios	Dynamics 365 Business Central live connector
P&L → Balance Sheet by mapping, Cash Flow from the delta	Multi-entity consolidation and multi-currency
Deviation calculated deterministically, explained by AI	Phases 1, 3 and 4 of the four-phase loop
Backtested accuracy before any claim of reliability	

THE EVALUATION CRITERION

Newvision has said, repeatedly and in writing, that transparency is what they are judging. The differentiator is not feature count — it is that every number can be explained, traced and defended live, including on figures they propose in the room. The architecture below is shaped around that single requirement.

SECTION 2

The mechanism, precisely

Rob's correction defines the model: a single blended variator is wrong because costs have a fixed and a variable component — a base fee plus usage, in his energy-cost example. So each cost category is split by regression against historical revenue, and the two halves behave differently in the plan.

Deriving the parameters – from actuals, never configured

For each cost category c , regress historical cost on historical revenue:

$$\text{Cost}_{c,t} = \alpha_c + \beta_c \times \text{Revenue}_t + \varepsilon$$

α_c – fixed component, absolute per year

β_c – variable rate, cost per unit of revenue

R^2_c – how well the split explains the history (surfaced, not hidden)

v_c – valorization rate: average year-over-year growth of the fixed part

Ordinary least squares over the historical window (5 years in the current model). Every parameter is a regression output over real Istwerten – nothing is entered as an assumption.

Projecting the plan

$$\text{PlanCost}_{c,t} = \alpha_c \times (1 + v_c)^{t-t_0} + \beta_c \times \text{PlanRevenue}_t$$

The fixed part valorizes forward on its own trend. The variable part follows planned revenue. This is the whole cascade – no separate cascade step is needed, because a change in revenue propagates through β automatically while α stays put. That is precisely the behaviour a blended variator got wrong.

Two special cases

- **Other costs / Depreciation** — confirmed as driven by investment levels, not estimated independently. Depreciation is therefore computed from an investment plan input on its own schedule, and only the remainder of Other costs takes the regression treatment.
- **Personnel costs** — Newvision runs at roughly 80% of total costs against an industry-typical 65%, and their own scenarios centre on workforce change. The regression still applies, but the fixed component here is the number people will interrogate hardest, so its derivation gets the most visible drill-down.

WHY THE THREE SCENARIOS ARE NEARLY FREE

Because everything downstream is a deterministic function of planned revenue, Best / Base / Worst require only three revenue paths. The full P&L, Balance Sheet and Cash Flow for each scenario fall out of the same engine — no separate scenario modelling, and no opportunity for the three to become inconsistent with each other.

Downstream chain – all deterministic

Planned P&L feeds the Balance Sheet through Newvision's own mapping (Rob confirmed the mapping is his, and is not an AI job). Cash Flow is then the delta between the planned P&L and the planned Balance Sheet. In the PoC this chain is config-driven: the mapping table is an input, seeded with a working example and clearly labelled as illustrative until Newvision supplies theirs.

SECTION 3

Where AI is, and where it is not

Rob's design philosophy is the governing rule: *"if all is determined by mathematics, we do not need AI — AI should analyze the business and the boundaries, challenges."* Three touchpoints, and nothing else.

STEP	HOW IT WORKS	AI?
Load actuals	Historical Istwerten per category, over the defined window	No
Fixed / variable split	OLS regression against revenue; α , β , R^2 per category	No — deterministic
Valorization rate	Average YoY growth of the fixed component	No — deterministic
Environmental scan the 54-position framework	AI investigates external factors and flags the material ones, with sources and written reasoning	Yes — touchpoint 1
Revenue proposal only when a factor is flagged	AI receives actuals, the valorized default, the external note and the control-table rules; proposes a value with written reasoning. A human reviews and confirms before anything is used	Yes — touchpoint 2
Cost projection & cascade	α , β applied to the confirmed revenue path	No — deterministic
P&L → Balance Sheet → Cash Flow	Mapping supplied by Newvision; cash flow from the delta	No — deterministic
Deviation calculation	Plan versus actual, arithmetic	No — deterministic
Deviation explanation	AI explains why a calculated deviation occurred, citing the contributing figures	Yes — touchpoint 3

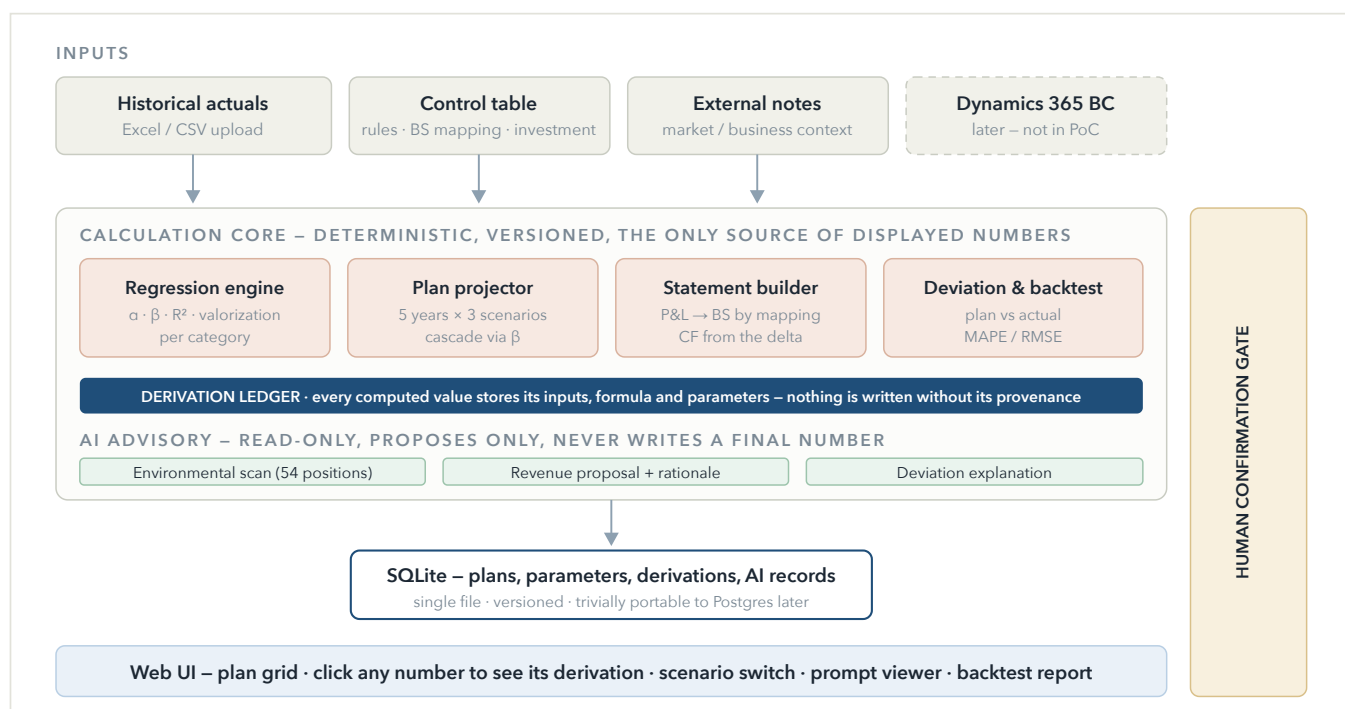
THREE HARD RULES, ENFORCED IN CODE

No AI write-access to any source system, ever — read-only and advisory. Every AI-produced value carries a written rationale and cannot exist as a bare number. And the literal prompt used at any AI step is stored and viewable in the UI, because being able to *show* the method beats being able to describe it.

SECTION 4

Architecture

Deliberately small. Three weeks does not permit infrastructure, and the data does not require it: five categories across ten years and three scenarios is a few thousand rows. Everything below runs as one process plus a static front-end, in a single container.



Rust = deterministic calculation. Green = the three AI touchpoints, which sit beside the core and propose into it, never inside it. Nothing an AI produces reaches a statement without passing the confirmation gate on the right.

COMPONENT	CHOICE	WHY THIS, FOR THREE WEEKS
Backend	Python + FastAPI	pandas/statsmodels do the regression natively; one language for maths and API.
Calculation	pandas + numpy (OLS)	The entire model is a few hundred lines. Pure functions, no framework, exhaustively testable.
Store	SQLite via SQLAlchemy	Data is tiny. Zero ops cost, and the ORM makes the later move to Postgres mechanical. Platform decision is still open — this commits us to nothing.
Front-end	React + Vite + Tailwind	Fast to generate, looks professional, and the click-to-trace interaction needs real components rather than a notebook UI.

COMPONENT	CHOICE	WHY THIS, FOR THREE WEEKS
AI	Claude API, three prompts	Prompt text, response, model version and timestamp stored with every proposal so the method can be shown, not just described.
Packaging	One Docker container	Runs on a laptop for the demo or on a server if they ask. No hosting decision required to proceed.

SECTION 5

Data model

Nine tables. The unusual one is `derivation` — it exists because traceability is the product, and retrofitting provenance is far harder than writing it as you compute.

category	(id, code, name, kind revenue cost, driver revenue investment)	
actual	(category_id, year, value, source_label)	Istwerten
parameter	(category_id, alpha, beta, r_squared, valorization_rate, window_from, window_to, computed_at, calc_version)	
scenario	(id, kind best base worst, label, created_by)	
plan_value	(scenario_id, category_id, year, value, path, derivation_id)	path: valorized ai_proposed cascaded
external_note	(id, category_id, year, text, author, created_at)	
ai_record	(id, touchpoint, prompt_text, response_text, rationale, model_version, proposed_value, status proposed confirmed rejected, confirmed_by, confirmed_at)	
derivation	(id, formula_text, inputs_json, parameters_json, computed_at)	
statement_line	(scenario_id, statement pl bs cf, line_code, year, value, mapping_ref, derivation_id)	

- Every `plan_value` and `statement_line` points at a `derivation`. A value without provenance cannot be inserted — enforced by a not-null foreign key, not by convention.
- `ai_record.status` starts at `proposed`. Only a human transition to `confirmed` lets the value enter `plan_value`. This is the confirmation gate, in one column.
- `parameter` rows are immutable and versioned: rerunning the regression on a wider window creates a new row, so historic plans remain reproducible.
- `actual.source_label` carries the "illustrative" flag until Newvision supplies real figures, and the UI renders it visibly — never silently.

SECTION 6

Traceability — the thing being judged

The requirement is that anyone can trace any figure back to its source in under a minute. In practice that means one interaction: click a number, get its full lineage.

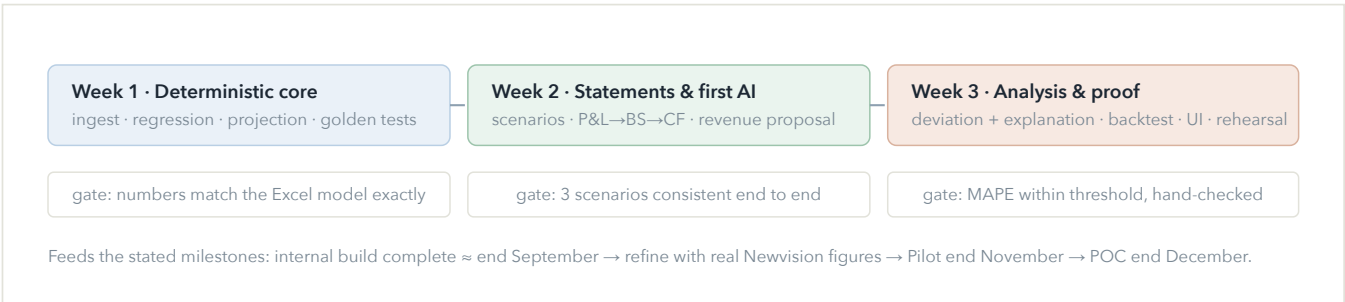
```
Personnel costs · 2028 · Base          4,812.4 k€
└ path: cascaded
└ formula:  $\alpha \times (1+v)^t + \beta \times \text{PlanRevenue}$ 
└ parameters:  $\alpha = 3,104.0$    $v = 2.8\%$    $\beta = 0.0910$    $R^2 = 0.94$ 
    └ regression over actuals 2021–2025 [show the 5 points and the fit]
└ inputs: PlanRevenue 2028 = 18,760.0 k€
    └ path: ai_proposed → confirmed by C. Andres, 14.10.2026
        └ external note: "major client contract ends Q2 2027"
        └ rationale: [full written reasoning]
        └ prompt: [the literal prompt text used]
└ source data: Istwerten_2021–2025.xlsx [ILLUSTRATIVE]
```

Two consequences worth stating plainly. First, this is why the derivation ledger is a first-class table rather than a logging afterthought. Second, it is what allows the live test that will decide the room: someone proposes a revenue figure on the spot, the plan recalculates, and every dependent number can be opened and defended immediately.

SECTION 7

The three-week plan

Fifteen working days, sequenced so that the deterministic core is provably correct before any AI is added — if week three is lost entirely, what remains is still a defensible planning engine.



Week 1 – the deterministic core

Goal: the engine reproduces the Excel reference model to the cent, with tests proving it.

DAY	WORK	DONE WHEN
1	Repo, container, schema, SQLAlchemy models, seed the five categories	Empty app runs; migrations apply
2	Ingest: Excel/CSV of actuals → actual rows, with source labelling and the illustrative flag	Reference workbook's actuals load cleanly
3	Regression engine: α , β , R^2 , valorization per category DETERMINISTIC	Parameters match hand-computed OLS on the same data
4	Plan projector: five-year projection, fixed valorized + variable on revenue	Projection matches the Excel model row for row

DAY	WORK	DONE WHEN
5	Derivation ledger wired through both; golden-case test suite from the workbook	Every number carries provenance; tests green

Week 2 – statements, scenarios, first AI touchpoint

Goal: three complete scenarios through to cash flow, and the revenue-proposal loop working end to end.

DAY	WORK	DONE WHEN
6	Scenario layer: Best/Base/Worst revenue paths driving the full model	Three scenarios, mutually consistent, one engine
7	Statement builder: P&L → Balance Sheet via config mapping DETERMINISTIC	BS balances; mapping visibly marked as illustrative
8	Cash Flow from the P&L/BS delta; depreciation from the investment input	CF ties to the movement in the BS
9	AI touchpoint: external note → revenue proposal + rationale AI ; prompt/response persisted	Proposal appears with its prompt viewable
10	Human confirmation gate: review, confirm/reject, cascade on confirm	Nothing enters the plan unconfirmed; rejection leaves no trace in figures

Week 3 – analysis, proof of accuracy, and the demo

Goal: the numbers are demonstrably right, and the story can be told live.

DAY	WORK	DONE WHEN
11	Backtest harness: train on early years, predict a known year, MAPE/RMSE per category	Report renders; thresholds evaluated per category
12	Deviation calculation DETERMINISTIC + AI explanation of the deviation AI	Explanation cites the actual contributing figures
13	UI: plan grid, scenario switch, click-to-trace panel, prompt viewer	Any number traceable in one click
14	Environmental scan surface: the 54-position structure, AI investigating a subset live AI	Scan runs on demand and writes into external notes
15	Hand-verification pass over every displayed figure; live-input rehearsal	Someone outside the build has checked the numbers by hand

The hand-verification pass is a deliverable, not slack. This project has already caught two real bugs that way — a misreferenced cell that silently swapped two values, and a contradictory personnel-cost assumption — and neither was caught by tests passing. If the schedule slips, cut scope from day 14, never from day 15.

SECTION 8

Building this with AI — where it helps, where it must not be trusted

Three weeks is realistic *because* of AI-assisted development, but only if the division of labour is deliberate.

AI ACCELERATES THIS WELL	VERIFY BY HAND, ALWAYS
Scaffolding: models, migrations, API routes, container setup	Every regression output, against hand-computed OLS
React components and the trace-panel UI	The projection arithmetic, against the Excel model
Excel/CSV parsers and their fixtures	Balance-sheet balancing and the cash-flow delta
Test case generation from the reference workbook	MAPE/RMSE computation and thresholds
Prompt drafting for the three touchpoints	Any figure that will appear on screen in front of Newvision

The rule that makes this safe is the one already in the Ledger: a model that runs without errors is not the same as a model that is correct. AI-generated code passes its own tests happily; what it cannot do is notice that a plausible-looking number is wrong. That judgement stays human, and it is budgeted for on day 15.

SECTION 9

Quality gates

GATE	CRITERION	WHEN
Reproduction	Engine output matches the Excel reference model exactly for Material costs, then all five categories	End of week 1
Formula integrity	No hardcoded results anywhere; changing any input visibly recalculates everything downstream	Continuous
Consistency	Three scenarios reconcile; BS balances; CF ties to BS movement	End of week 2
Backtested accuracy	MAPE/RMSE reported per category and horizon against the 5–8% governance threshold — and stated honestly where a category misses it	Week 3, day 11
Provenance	Every displayed figure traceable in one click; every AI value carries prompt, rationale and confirmer	Week 3, day 13

GATE	CRITERION	WHEN
Hand verification	An independent pass over every on-screen number against hand calculation	Week 3, day 15

ON THE 5-8% THRESHOLD

This is achievable for stable cost categories on a one-year-ahead backtest, and considerably harder for Revenue over longer horizons. Report it per category *and* per horizon rather than as one headline number, and say plainly where it is not met. Presenting a missed threshold honestly is survivable; presenting an averaged figure that hides it is not.

SECTION 10

What is cut, and what could go wrong

Deliberately out of the three weeks

- **CUT** Live Dynamics 365 Business Central connector — file ingest only. Real figures are not confirmed yet, so a connector would be built against assumptions.
- **CUT** Multi-entity consolidation and multi-currency. Real, but Step 1 is total-company single-currency.
- **CUT** Full 54-position scan depth — the structure is shown and a subset investigated live. Depth is still an open question with Newvision anyway.
- **CUT** Authentication and multi-user roles. Single-operator demo; the confirmation gate records a named confirmer without needing an identity system.
- **CUT** Anything in Step 2 — HR, Resource, Portfolio Planning — regardless of how easy it looks to bolt on.

Risks, in order of likelihood

RISK	MITIGATION
Real actuals arrive late or not before the build ends	Build entirely on the reference workbook's data, labelled illustrative throughout. Swapping in real figures is a data load, not a rebuild — that is the point of deriving every parameter rather than configuring it.
Balance-sheet mapping unavailable	Ship a worked example mapping, visibly marked as ours, and structure it as config so Newvision's version drops in unchanged.
A category misses the accuracy threshold	Report per category, show the fit visually, and let the R^2 tell the story. A weak fit on one category with an honest explanation is more credible than a suspiciously uniform result.
Platform decision arrives mid-build	SQLite + SQLAlchemy + one container means the answer changes deployment, not architecture.

RISK	MITIGATION
Scope creep from adjacent asks	Step 2 stays Step 2. The Ledger is explicit, and precision on a narrow scope is the stated strategy.

Open items that would change this plan

1. Which specific accounts map to each of the five categories, and the real historical figures behind them.
2. Newvision's P&L → Balance Sheet mapping.
3. Depth and scope of the environmental scan — and whether findings are shown to Newvision directly or summarised first.
4. Platform and hosting.

Source hierarchy: Team Briefing – Complete Requirements Ledger governs. Development Team Briefing supplies background; where the two differ on mechanism, the Ledger's Fixed + Variable regression supersedes the earlier single-variator description.

AI-Supported Planning · PoC Architecture & Implementation Plan · PerfektWerk for Newvision · September 2026.